Alphabrain Platform Technical Report

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https://github.com/AlphaBrainGroup/AlphaBrain
https://www.alphabrain-platform.com/

Full Author List in Contributions and Acknowledgements

Abstract

We present Alphabrain Platform, a unified platform for building generalist embodied intelligence. Alphabrain Platform integrates large-scale vision–language pretraining, robot trajectory learning, and efficient downstream adaptation into a single, end-to-end recipe. The platform produces Alphabrain, a vision–language–action (VLA) model that follows natural-language instructions and generalizes to novel objects, environments, and tasks. Alphabrain Platform is designed for robustness on long-horizon and dexterous manipulation, and supports rapid, cost-effective fine-tuning from a small number of human demonstrations. Through extensive real-world experiments, we show that Alphabrain Platform delivers strong instruction following, reliable long-horizon execution, and sample-efficient adaptation. We hope Alphabrain Platform serves as a practical foundation for deploying generalist robots in everyday settings.

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1 Introduction

The pursuit of generalist robots capable of assisting humans with everyday tasks has been a long-standing goal in robotics. A central challenge stems from the diversity of the real world, which demands policies that generalize across a wide range of objects, scenes, and instructions. Recent advances in Vision–Language–Action (VLA) models [1] have paved a promising path toward intelligent generalist policies.

figures/overview.pdf — system overview goes here

Figure 1 Overview. Alphabrain Platform learns from vision–language data, robot trajectory data, and human trajectory data, and performs dexterous, long-horizon tasks with strong robustness and generalization.

In this report, we introduce Alphabrain Platform, …*(Replace this paragraph with your contribution summary; see Figure 1.)*

We make the following contributions:

- A unified training recipe combining web-scale vision–language data with robot and human trajectory data.
- Alphabrain, a VLA model that strictly follows instructions and generalizes to novel settings.
- Extensive real-world experiments demonstrating robust long-horizon and dexterous capabilities.

2 The Alphabrain Model

Alphabrain is an end-to-end vision–language–action model π_θ. It takes as input a language instruction l, multi-view observations $\mathbf{o}_t$, and the robot state $\mathbf{s}_t$, and predicts an action chunk $\mathbf{a}_t = \pi_\theta(l, \mathbf{o}_t, \mathbf{s}_t)$.

2.1 Architecture

Describe the backbone, the action head, and any normalization choices here.

2.2 Action Prediction

We supervise action prediction with a flow-matching objective:

$$L_{\text{action}}(\theta) = \mathbb{E}_{\{\mathbf{a}_t, \mathbf{o}_t, \mathbf{s}_t, l\} \sim \mathcal{D}} \left[\left\| \mathbf{v}_\theta(l, \mathbf{o}_t, \mathbf{s}_t, \mathbf{a}_t^\tau) - \mathbf{u}(\mathbf{a}_t^\tau \mid \mathbf{a}_t) \right\|^2 \right], \tag{1}$$

where $\tau \sim \mathcal{U}(0, 1)$ is the flow-matching timestep.

Table 1 **Main results.** Average task progress (%) across benchmarks.

Method	Pick-and-Place	Table Bussing	Cloth Manip.
Baseline	00.0	00.0	00.0
Alphabrain Platform	**00.0**	**00.0**	**00.0**

3 Training Recipe

We train Alphabrain on a mixture of data sources, including robot trajectory data for imitation learning, web-scale vision–language data for co-training, and human trajectory data for few-shot generalization.

3.1 Imitation Learning with Robot Trajectory Data

We maximize the log-likelihood of the policy on expert demonstrations $\mathcal{D}$:

$$\max_{\theta} \; \mathbb{E}_{\{\mathbf{a}_t, \mathbf{o}_t, \mathbf{s}_t, l\} \sim \mathcal{D}} \Big[\log \pi_\theta(\mathbf{a}_t \mid \mathbf{o}_t, \mathbf{s}_t, l) \Big]. \tag{2}$$

3.2 Co-training with Vision–Language Data

Explain the co-training mixture and weighting here.

4 Experiments

Summarize the experimental setup, baselines, and findings here, referencing Table 1.

5 Conclusion

We introduced Alphabrain Platform, a unified platform for generalist embodied intelligence, and Alphabrain, the VLA model it produces. We hope it serves as a step toward broadly capable, reliable robots.

Contributions and Acknowledgements

References

[1] Jane Doe and John Smith. A vision–language–action model for generalist robots. *arXiv preprint arXiv:2500.00000*, 2025.